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qwx762

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Test Report

Automated Testing and Results

This project was validated with automated functional tests to confirm that core requirements work correctly. Testing was run against the API layer using a separate test database.

Test Setup

- Test runner: Vitest
- Environment: Node test environment with isolated SQLite database
- Test files:
 - api.functional.test.ts
 - setup.ts
 - vitest.config.ts
- Run command: npm run test:report

What Was Tested

- Account creation and password hashing
- Duplicate signup prevention
- Sign-in privilege enforcement for creating channels and posts
- Input validation for post creation
- Nested replies using parentReplyId
- Voting rules:
 - one vote per user per target
 - change vote value

- remove vote using neutral state
- Admin moderation permissions:
 - non-admin cannot delete channel
 - admin cannot delete another admin
- Screenshot upload validation:
 - unsupported MIME rejected
 - PNG accepted and stored
- Search behavior:
 - author search returns author displayName for posts and replies

Automated Test Cases Executed

1. Signup creates user with hashed password
2. Signup blocks duplicate usernames
3. Creating channel requires sign in
4. Signed-in user can create channel
5. Creating post requires sign in
6. Creating post validates required fields
7. Replies support nested threading via parentReplyId
8. Voting enforces one vote per user per target
9. Neutral vote removes existing vote
10. Non-admin cannot delete channel
11. Admin cannot delete another admin account
12. Upload rejects unsupported MIME type
13. Upload accepts PNG and stores attachment metadata
14. Author search returns posts/replies with author displayName

Proof of Results (Terminal Output)

The following output was produced by `npm run test:report`:

- ✓ tests/api.functional.test.ts (14 tests)
- ✓ Core functional API requirements (14)

Test Files 1 passed (1)

Tests 14 passed (14)

Duration 4.73s

```
PS C:\Users\Carter\Documents\CMPT353\qwx762_Project> npm run test:report
> qwx762-project@0.1.0 test:report
> vitest run --reporter=verbose

RUN v3.2.4 C:/Users/Carter/Documents/CMPT353/qwx762_Project

Loaded Prisma config from prisma.config.ts.

Prisma schema loaded from prisma/schema.prisma.
Datasource "db": SQLite database "test.db" at "file:./test.db"

SQLite database test.db created at file:./test.db

The SQLite database "test.db" at "file:./test.db" was successfully reset.

Your database is now in sync with your Prisma schema. Done in 38ms

✓ tests/api.functional.test.ts (14 tests) 4031ms
  ✓ Core functional API requirements (14)
    ✓ 1) signup creates user with hashed password 289ms
    ✓ 2) signup blocks duplicate usernames 81ms
    ✓ 3) creating channel requires sign in 11ms
    ✓ 4) signed-in user can create channel 91ms
    ✓ 5) creating post requires sign in 18ms
    ✓ 6) creating post validates required fields 87ms
    ✓ 7) replies support nested threading via parentReplyId 138ms
    ✓ 8) voting enforces one vote per user/target (upsert update) 144ms
    ✓ 9) vote value 0 removes existing vote (neutral state) 182ms
    ✓ 10) non-admin cannot delete channel 171ms
    ✓ 11) admin cannot delete another admin account 210ms
    ✓ 12) upload rejects unsupported mime type 94ms
    ✓ 13) upload accepts png and stores attachment metadata 113ms
    ✓ 14) author search returns posts/replies with author displayName 136ms

Test Files 1 passed (1)
Tests 14 passed (14)
Start at 16:52:50
Duration 4.73s (transform 171ms, setup 27ms, collect 400ms, tests 4.03s, environment 0ms, prepare 102ms)
```

Result Summary

All 14 automated test cases passed. This provides reproducible evidence that the main functional requirements are implemented and working as expected.